Snowy Plover Game - Options For Decisions and Information for User

Game Choices

Bird Description/Summary:
The Western Snowy Plover nests on sandy beaches during the summer. They nest in small, camouflaged dips on the open sand, decorated with small bones, pebbles, and shell fragments. They feed on a variety of invertebrates such as beetles, flies, snails, juvenile mole crabs, and brine fly larvae. The snowy plover struggles to breed due to a combination of human disturbance, predators, harsh weather, and a loss of nesting habitat in the sandy dunes. Could you survive a nesting season as a snowy plover?

*Note: This summary might be too long, lets test and see if the length needs to be cut down

*****Note: Bad Choices are based on Educated Guesses and should be checked with the Oregon Aquarium team from those educated on Snowy Plovers.

Choose Beach:Mentioned in Summary: (Yes)
Good Option:
Sandy Beach
Sandy Dune with Plants

Bad Options:
Rocky Cliff
Grass Field
Coastal Town

Choose Food: Mentioned in Summary: (Yes)
Good Options:
Juvenile mole crab
Brine fly larvae
Snails
Flies
Beetles

Bad Options:
Rocks
Bull Kelp
Sea Lettuce (It's a kind of kelp)

Fish
French Fries

Choose Nest Material/Camouflage: Mentioned in Summary:(YES)
Good Options:
Small Bones
Pebbles
Shell Fragments
Seaweed
Feathers

Bad Options: (They will occasionally use white plastics so while these aren't ideal, the birds will use these items to their advantage)
Colorful Fast Food Bags (Flashy to predators)
Shiny toy jewels (Flashy to predators)
(Anything that would attract predators)

Quick text for after nest material screen: Choosing nest materials that insulate warmth and keep the nest hidden is essential for keeping eggs safe!

Avoid Danger: Mentioned in Summary: (No)
The beach has many different inhabitants, what should we avoid when choosing our nest location:

Avoid:
Dogs Offleash
Ravens (Eats plover eggs and chicks) (refer to Hiding in Plain Sight: Western Snowy Plovers 28:49)
Recreational Vehicles (dune buggies, ebikes)
Drones

No Need to Avoid:
Native Vegetation (examples: Pink Sand Verbena, seashore lupine, beach knotweed)
Other Shorebirds (When near the tide they are fine, Snowy Plovers are territorial so check with bird biologists if these birds in feeding areas would still cause problems. Examples: Black Oystercatchers, Sandpipers, Killdeer) (Other birds that strictly eat fish such as pelicans might be fine too)

Text for after the threat screen: Plovers will flee from a nest when they detect a threat nearby. During this flee they will cry out and drag their wings to act as an easy target, bringing the predator away from the nest!

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Notes:

Additional Options For Decisions: Nest Location on the beach in terms of how far from the water the nest is?

Challenge: Creating Bad Options

Question: Should we mention non native european beachgrass and hint at the harm they do to the Oregon sand dunes? Would adding text after the beach and food screens be too redundant? Where are inaccuracies in these decisions, questions, and information?

Comment: For referencing our sources and other contributions, we should include a credits button on the main screen for a user to select to see where research was pulled from and who contributed to the project.

Comment from Reviewer: If there are many choices for each decision(good) maybe we do an even number (4) choices per screen, two good two bad.

Game Results:

Great Success:

Congratulations, your snowy plover was highly successful this nesting season! Great job at making informed decisions and keeping your snowy plover safe!
The Snowy Plover population used to only be estimated at 35-50 birds in Oregon during the early 1990's. With conservation efforts over the years, as of 2026 there are an estimated 3018 Western Snowy Plovers along the West Coast of the United States!
Make sure to recreate safely on our beaches and stay out of Snowy Plover Management Areas when signage is posted so that we can all have fun together!

Mild Success:

Congratulations, your snowy plover made it through this nesting season! Ensuring the conservation of these birds is an important and delicate mission that helps us maintain a thriving coastal ecosystem!
The Snowy Plover population used to only be estimated at 35-50 birds in Oregon during the early 1990's. With conservation efforts over the years, as of 2026 there are an estimated 3018 Western Snowy Plovers along the West Coast of the United States!

Make sure to recreate safely on our beaches and stay out of Snowy Plover Management Areas when signage is posted so that we can all have fun together!

Failure:

Your snowy plover did not get any eggs this nesting season. If you want to play again, make sure to read the summary about the snowy plover to increase your chances of success!
The Snowy Plover population used to only be estimated at 35-50 birds in Oregon during the early 1990's. With conservation efforts over the years, as of 2026 there are an estimated 3018 Western Snowy Plovers along the West Coast of the United States!
Make sure to recreate safely on our beaches and stay out of Snowy Plover Management Areas when signage is posted so that we can all have fun together!

Resources

https://birdallianceoregon.org/our-work/protect/habitat-and-wildlife/imperiled-species/western-snowy-plover/ (Most information is from here, including historical population count facts used in the end game text)

https://www.fws.gov/story/2026-04/count-worth-waiting (Only used for current population count information)

https://www.allaboutbirds.org/guide/Snowy_Plover/lifehistory (Good diet and behavior Information)

https://www.montereybayaquarium.org/animals-the-ocean/animals-a-to-z/western-snowy-plover

https://www.youtube.com/watch?v=eAcFh8P2PnA (Very helpful talk on these birds, lots of information and good pictures!)

https://www.oregon.gov/oprd/PCB/Documents/WSP-HCP_08182010-web.pdf Habitat Conservation Plan for Snowy Plover

https://www.nrcs.usda.gov/plantmaterials/orpmcsr15058.pdf Native Plant Materials for Coastal Dune

https://www.youtube.com/watch?v=SmiNol6rgcE Good Nest Choosing Information (is 15 years old, might be too dated)

https://papers.risingsea.net/rolling-easements-2-2-1.html (Beach Zones, Important to know what zones matter for nesting and feeding (these birds are territorial during nesting/breeding!)

https://thewatershedproject.org/whats-in-your-watershed-western-snowy-plover/